VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Configure a ZPF

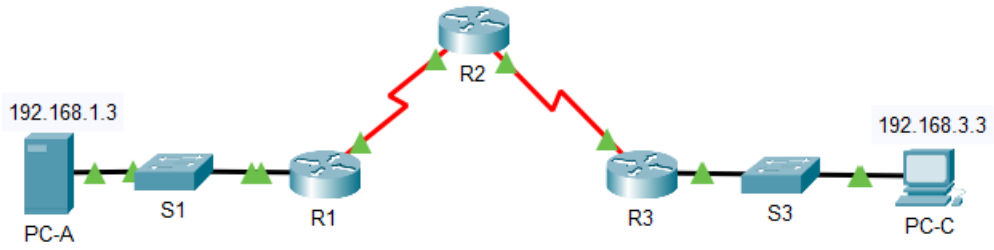
Prepared by:
Kudzaishe Majeza
Junior Network Engineer – VxLEARN Networks

Mentor:
Titus Majeza
Senior Network Engineer

Date: 7 August 2026

1. Objectives

- Verify connectivity among devices before firewall configuration.
- Configure a zone-based policy firewall (ZPF) on R3.
- Verify ZPF firewall functionality using ping, SSH, and a web browser.



2. Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
R1	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
R2	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
R3	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

3. Background / Scenario

Zone-Based Policy Firewalls (ZPFs) are the latest development in the evolution of Cisco firewall technologies. In this activity, a basic ZPF is configured on edge router R3 that allows internal hosts access to external resources while blocking external hosts from accessing internal resources. Firewall functionality is then verified from both internal and external hosts.

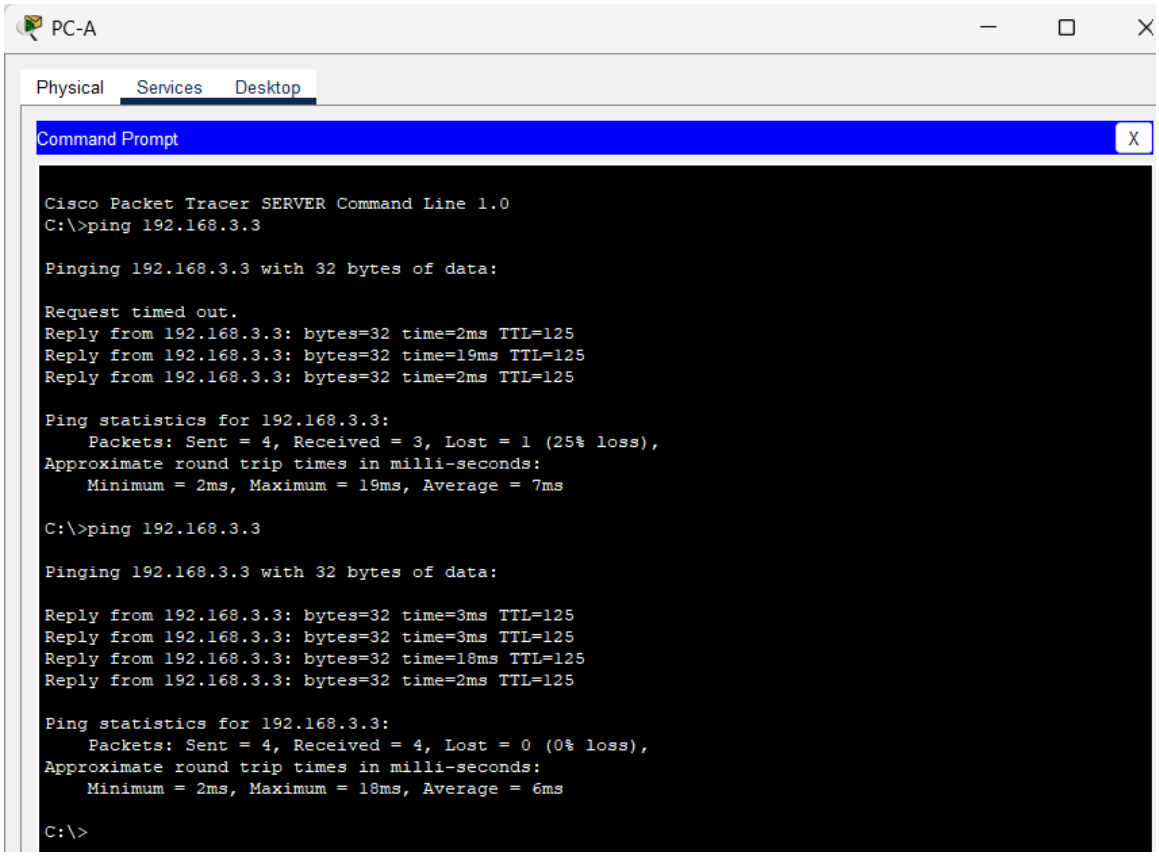
The routers were pre-configured with the following: console password ciscoconpa55, vty password ciscovtypa55, enable password ciscoenpa55, host names and IP addressing, a local username/password of Admin / Adminpa55, and static routing.

4. Part 1 – Verify Basic Network Connectivity

Connectivity is verified before the firewall is configured, to confirm that any later failures are caused by the firewall policy and not by an underlying routing or addressing problem.

Step 1: Ping PC-A from PC-C

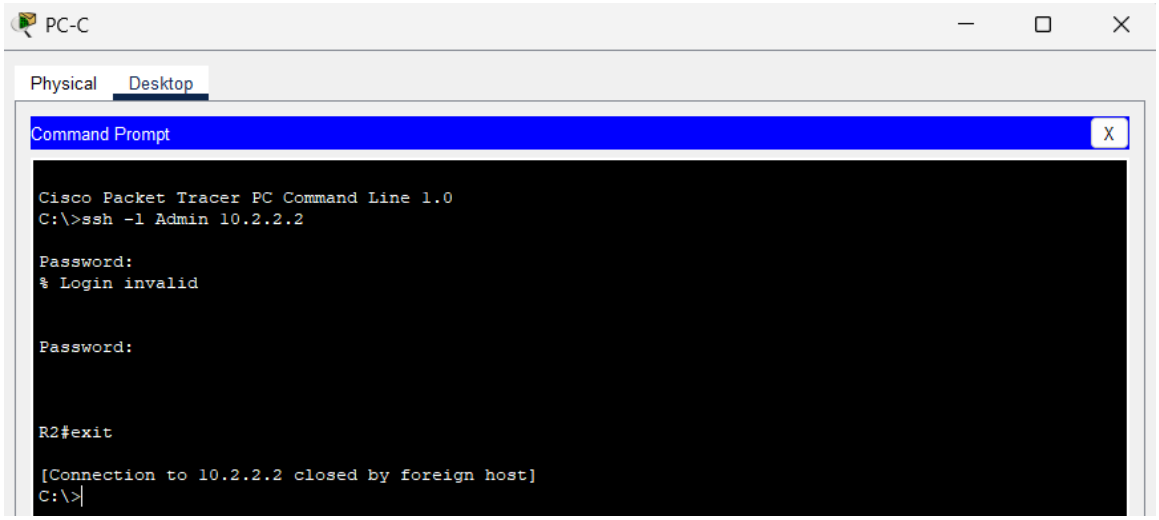
From the PC-A command prompt, ping PC-C at 192.168.3.3. The ping should succeed.



Step 2: Access R2 using SSH

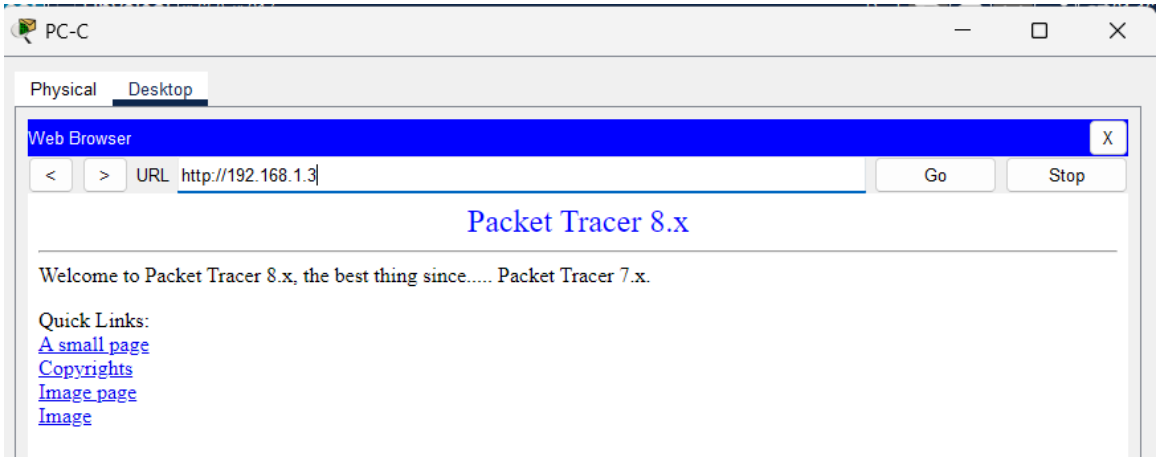
```
C:\> ssh -l Admin 10.2.2.2
```

Log in with username Admin and password Adminpa55, then exit the SSH session.



Step 3: Open a web browser from PC-C to the PC-A server

On PC-C's Desktop tab, open the Web Browser application and enter 192.168.1.3 as the URL. The Packet Tracer welcome page should display.



5. Part 2 – Create the Firewall Zones on R3

Every ZPF configuration begins by creating the security zones. Names must match exactly, since later steps (class-maps, zone-pairs, interface assignments) all reference these names.

Step 1: Create the internal zone

```
R3(config)# zone security IN-ZONE
R3(config-sec-zone)# exit
```

Step 2: Create the external zone

```
R3(config)# zone security OUT-ZONE
R3(config-sec-zone)# exit
```

6. Part 3 – Identify Traffic Using a Class-Map

Step 1: Create an ACL defining internal traffic

```
R3(config)# access-list 101 permit ip 192.168.3.0 0.0.0.255 any
```

Step 2: Create a class-map referencing the ACL

```
R3(config)# class-map type inspect match-all IN-NET-CLASS-MAP
R3(config-cmap)# match access-group 101
R3(config-cmap)# exit
```

7. Part 4 – Specify Firewall Policies

Step 1: Create the policy-map

```
R3(config)# policy-map type inspect IN-2-OUT-PMAP
```

Step 2: Reference the class-map

```
R3(config-pmap)# class type inspect IN-NET-CLASS-MAP
```

Step 3: Specify the inspect action

```
R3(config-pmap-c)# inspect

%No specific protocol configured in class IN-NET-CLASS-MAP for inspection.
All protocols will be inspected.

R3(config-pmap-c)# exit
R3(config-pmap)# exit
```

The inspect command invokes stateful (context-based) access control. The other available actions are pass (stateless forwarding) and drop (silently discard).

8. Part 5 – Apply Firewall Policies

Step 1: Create the zone pair

```
R3(config)# zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-
ZONE
```

Step 2: Attach the policy-map to the zone pair

```
R3(config-sec-zone-pair)# service-policy type inspect IN-2-OUT-PMAP
R3(config-sec-zone-pair)# exit
R3(config)#
```

Step 3: Assign interfaces to their security zones

```
R3(config)# interface g0/1
R3(config-if)# zone-member security IN-ZONE
R3(config-if)# exit
R3(config)# interface s0/0/1
R3(config-if)# zone-member security OUT-ZONE
R3(config-if)# exit
```

Step 4: Save the configuration

```
R3# copy running-config startup-config
```

9. Part 6 – Test Firewall Functionality: IN-ZONE → OUT-ZONE

Confirms that internal hosts can still reach external resources after the ZPF is applied.

Step 1: Ping PC-A from PC-C

From PC-C, ping PC-A at 192.168.1.3. The ping should succeed.

Step 2: SSH from PC-C to R2, and inspect the session on R3

From PC-C, SSH to R2 at 10.2.2.2 (username Admin, password Adminpa55). While the session is active, run the verification command on R3:

```
R3# show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
  Zone-pair: IN-2-OUT-ZPAIR
  Service-policy inspect : IN-2-OUT-PMAP
    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect
        Number of Established Sessions = 1
        Established Sessions
          Session 175216232 (192.168.3.3:1028)=>(10.2.2.2:22) tcp
SIS_OPEN/TCP_ESTAB
            Created 00:00:25, Last heard 00:00:20
            Bytes sent (initiator:responder) [1195:1256]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
```

What is the source IP address and port number?

192.168.3.3:1028 (port 1028 is a random source port)

What is the destination IP address and port number?

10.2.2.2:22 (SSH uses port 22)

Step 3: Exit the SSH session

From PC-C, exit the SSH session and close the command prompt window.

Step 4: Open a web browser from PC-C to PC-A, and inspect the session on R3

Navigate to 192.168.1.3 in a browser on PC-C. The HTTP session should succeed. While active, run the same verification command on R3 again:

```
R3# show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
  Zone-pair: IN-2-OUT-ZPAIR
  Service-policy inspect : IN-2-OUT-PMAP
    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect
        Number of Established Sessions = 1
        Established Sessions
          Session 565266624 (192.168.3.3:1031)=>(192.168.1.3:80) tcp
SIS_OPEN/TCP_ESTAB
            Created 00:00:01, Last heard 00:00:01
            Bytes sent (initiator:responder) [284:552]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
```

What is the source IP address and port number?

192.168.3.3:1031 (port 1031 is a random source port)

What is the destination IP address and port number?

192.168.1.3:80 (HTTP web traffic uses port 80)

Step 5: Close the browser

Close the browser on PC-C.

10. Part 7 – Test Firewall Functionality: OUT-ZONE → IN-ZONE

Confirms that external hosts cannot initiate connections into the internal network after the ZPF is applied.

Step 1: Ping PC-C from PC-A

From PC-A, ping PC-C at 192.168.3.3. The ping should fail.

Step 2: Ping PC-C from R2

From R2, ping PC-C at 192.168.3.3. The ping should fail.

Step 3: Check results

Completion percentage should read 100%. Click Check Results to confirm.

11. Final Device Configuration – R3

```
zone security IN-ZONE
zone security OUT-ZONE

access-list 101 permit ip 192.168.3.0 0.0.0.255 any

class-map type inspect match-all IN-NET-CLASS-MAP
 match access-group 101

policy-map type inspect IN-2-OUT-PMAP
 class type inspect IN-NET-CLASS-MAP
  inspect

zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE
 service-policy type inspect IN-2-OUT-PMAP

interface GigabitEthernet0/1
 zone-member security IN-ZONE

interface Serial0/0/1
 zone-member security OUT-ZONE
```

12. Analysis / Reflection

1. Why does only one zone-pair need to be configured, instead of one in each direction?

The inspect action is stateful: when IN-ZONE → OUT-ZONE traffic is inspected, R3 automatically tracks that session and permits its matching return traffic back through, without needing a separate OUT-ZONE → IN-ZONE policy. A second zone-pair is only needed if traffic has to be newly initiated from OUT-ZONE toward IN-ZONE, which this lab intentionally avoids.

2. Why does the ping from PC-A and R2 to PC-C fail, even though no explicit deny rule was configured?

ZPF defaults to deny-by-default between any two zones that don't have a zone-pair defined. Since only IN-2-OUT-ZPAIR (IN-ZONE source, OUT-ZONE destination) exists, any traffic entering from OUT-ZONE toward IN-ZONE that is not part of an already-inspected session has no matching zone-pair and is dropped automatically — this is the intended security behavior, not a configuration gap.

3. Why use match-all instead of match-any on the class-map, given there's only one match statement?

With a single match condition the two behave identically, but match-all is the more defensively correct choice: if a second match statement were added later (e.g., restricting by protocol too), match-all would require both conditions to be true, while match-any would loosen the policy by accepting either one alone.

4. What would happen if the inspect action were replaced with pass on the policy-map?

pass is stateless — it would forward IN-ZONE → OUT-ZONE traffic, but it would not automatically permit the return traffic, since pass does not track session state. A second zone-pair and policy would then be required for OUT-ZONE → IN-ZONE just to let replies back in, and the firewall would lose its ability to distinguish a real reply from an unsolicited new connection.

13. Conclusion

This lab applied the full five-step Zone-Based Policy Firewall configuration process — create zones, identify traffic with a class-map, define an action with a policy-map, bind a zone-pair to that policy, and assign zones to interfaces — to a real two-router topology. Verification with show policy-map type inspect zone-pair sessions confirmed the firewall was not just configured but actively tracking live SSH and HTTP sessions, while the IN-ZONE → OUT-ZONE/OUT-ZONE → IN-ZONE ping tests confirmed the core ZPF principle in action: internal hosts can always initiate outbound connections, but external hosts can never initiate inbound ones unless a matching zone-pair policy explicitly allows it.